EMCH 290-001-EE#3 Due at 11:59 on Tuesday, October 4, 2022

Points Possible: 100

1. **Question 1(25 Points)**

A rigid, well-insulated tank contains a two-phase mixture of ammonia with 0.005 m^3 of saturated liquid and 1.5 m^3 of saturated vapor, initially at p_1 = 5 bar. A paddle wheel stirs the mixture until only saturated vapor at higher pressure, p_2, remains in the tank. Kinetic and potential energy effects are negligible.

   a. Determine p_2, in bar.
   b. Determine the amount of energy transfer by work, in kJ.

2. **Question 2(25 Points)**

A piston–cylinder assembly contains 18 kg of ammonia, initially at 100 kPa and -10°C. The ammonia is compressed to a pressure of 550 kPa. During the process, the pressure and specific volume are related by $pv^{1.1} = constant$.

   a. Determine the work, in kJ.
   b. Determine the heat transfer, in kJ.

3. **Question 3(25 Points)**

A piston–cylinder assembly contains 0.25 kg of air initially at a pressure of 2 bar and a temperature of 200°C. The air is heated at constant pressure until its volume is doubled. Assume the ideal gas model with constant specific heat ratio, $k = 1.4$.

   a. Determine Initial Volume.
   b. Determine the work, in kJ
   c. Determine the heat transfer, in kJ

4. **Question 4(25 Points)**

Air is compressed in a piston–cylinder assembly from p_1 = 0.7 bar, T_1 = 275 K, V_1 = 0.25 m^3 to a final volume of V_2 = 0.03 m^3 in a process described by $pv^{1.25}$ = constant. Assume ideal gas behavior and neglect kinetic and potential energy effects. Using constant specific heats evaluated at T_1, determine the work and the heat transfer, in kJ.

   a. Determine the work, in kJ
   b. Determine the heat transfer, in kJ

1a) $V_{1f} = 0.005 m^3$ $p_1 = 5bar$

$V_{1g} = 1.5 m^3$

$v_{1g} = 0.2503$ $v_{1f} = 1.58\times10^{-3}$

$m_{1f} = \frac{V_{1f}}{v_{1f}} = \frac{0.005}{1.58\times10^{-3}} = 3.16 \text{ kg}$

$m_{1g} = \frac{V_{1g}}{v_{1g}} = \frac{1.5}{0.2503} = 5.99 \text{ kg}$

$v_2 = v_1 = \frac{V_{1f}+V_{1g}}{m_{1g}+m_{1f}} = \frac{1.5+0.005}{5.99+3.16} = 0.1645 \ m^3/kg$

$P_2 = \frac{v_{g7}-v_2}{v_{g7}-v_{g8}}(7-8)+8 = \frac{0.1815-0.1643}{0.1815-0.1596}(-1)+8 = \boxed{7.215 \text{ bar}}$

1b) $u_1 = \frac{v_{1g}-v_1}{v_{1g}-v_{1f}}(u_{1g}-u_{1f})+u_{1f}$

$= \frac{0.2503-0.1643}{0.2503-1.58\times10^{-3}}(1316.12-170.55)+170.55$

$= \frac{.086}{.24872}(1145.57)+170.55 = 566.65 \ \frac{kJ}{kg}$

$u_2 = \frac{v_7-v_2}{v_7-v_8}(u_7-u_8)+u_8$

$u_2 = \frac{0.1815-0.1643}{0.1815-0.1596}(1328.12-170.55)+1330.64$

$= \frac{0.0172}{0.0219}(1157.57)+1330.64 = 1328.598 \frac{kJ}{kg}$

$U_1 = m\cdot u_1 = (5.993+3.1645)(566.65) = 5189 kJ$

$\Delta U = U_2 - U_1 = 12166.6 - 5189 = 6977.6$

$\Delta U + \cancel{KE} + PE = \cancel{Q} - W$

$\boxed{W = -6977.6 kJ}$

2a) $p_1 = 100kpa$ $p_2 = 550kpa$

$T_1 = -10°C$

$v_1 = 1.2621 \ m^3/kg$

$p_2 v_2^n = p_2 v_2^n$

$v_2 = \left(\frac{p_1 v_1^n}{p_2}\right)^{1/n} = \left(\frac{100\cdot1.2621^{1.1}}{550}\right)^{\frac{1}{1.1}}$

$v_2 = 0.2679 \ m^3/kg$

$W = \int_1^2 p\,dV = \frac{m(p_2v_2 - p_1v_1)}{1-n}$

$= \frac{18((550\cdot0.2679)-(100\cdot1.2621))}{1-1.1}$

$= \frac{18(147.345-126.21)}{-.1}$

$\boxed{W = -3804.3 kJ}$

2b) $u_1 = 1324.33$ $u_2 = \frac{v_{50}-v_2}{v_{50}-v_{40}}(u_{50}-u_{40})+u_{40}$

$= \frac{0.27454-0.26794}{0.27454-0.26441}(1408.53-1389.64)+1389.64$

$= \frac{0.0066}{0.01013}(18.89)+1389.64 = 1401.95 \frac{kJ}{kg}$

$\Delta U + \underbrace{\cancel{KE}}_{=0} + \underbrace{\cancel{PE}}_{=0} = Q - W$ $\Delta U = m(u_2-u_1)$

$= 18(1401.95 - 1324.33) = 1397.16$

$Q = \Delta U + W$

$Q = 1397.16 - 3804.3$

$\boxed{Q = -2407.14 \ kJ/kg}$

3a) 25 kg air $P_1 = 2\text{ bar}$ $T_1 = 200°C$

$k = 1.4$

$R = \frac{\bar{R}}{M} = \frac{8.314 \frac{J}{mol \cdot K}}{28.97 \frac{kg}{kmol}} \times 1000 \frac{g}{kg} = 286.99 \frac{J}{kg \cdot K}$

$V = \frac{mRT}{P} = \frac{(.25)(286.99)(200+273)}{2\times10^5} = \frac{33936.57}{200000}$

$\boxed{V = 0.1697\ m^3}$

3b) $W = p(V_2 - V_1)$ $V_2 = 2V_1$

$W = p(2V_1 - V_1)$

$W = p(V_1) = \frac{(2\times10^5)(0.1697\ m^3)}{1000} = \boxed{33.94\ kJ}$

3c) $C_v = \frac{R}{k-1}$ $m(T_2 - T_1) = \frac{P}{R}(V_2 - V_1) = \frac{P}{R}V_1$

$U = \left(\frac{P}{\cancel{R}}V_1\right)\left(\frac{\cancel{R}}{k-1}\right) = \frac{PV_1}{k-1}$

$U + \cancel{KE} + \cancel{PE} = Q - W$

$Q = \frac{PV}{k-1} + W = \frac{(200)(0.1697)}{1.4-1} + 33.94$

$\boxed{Q = 118.79\ kJ}$

4a) $P_1 = 0.7\text{ bar}$

$T_1 = 275K$

$V_1 = 0.25 m^3$ $V_2 = 0.03 m^3$

$W = \frac{P_2V_2 - P_1V_1}{1-n}$

$P_2 = \frac{P_1V_1^n}{V_2^n} = 0.7\left(\frac{.25}{0.03}\right)^{1.25} = 9.91\text{ bar}$

$W = \frac{(9.91\times10^5)(0.03) - (0.7\times10^5)(0.25)}{1-1.25}$

$W = \frac{29730 - 17500}{-0.25} = \frac{48920\ J}{1000} = \boxed{-48.92\ kJ}$

4b) $U + \cancel{KE} + \cancel{PE} = Q - W$

$Q = U + W = m(U_2 - U_1) + W$

$R = \frac{\bar{R}}{M} = \frac{8.314}{28.97} \cdot 1000 = 287 \frac{J}{kg \cdot K}$

$m = \frac{PV}{RT} = \frac{(7\times10^5)(0.25)}{(287)(275)} = 0.2217\text{ kg}$

$T_2 = \frac{PV}{mR} = \frac{(9.91\times10^5)(0.03)}{(0.2217)(287)} = 467.25\ K$

$m(U_2 - U_1) = mC_v(T_2 - T_1)$

C_v @ 275K = 0.717

$Q = mC_v(T_2 - T_1) + W$

$= (0.2217)(0.717)(467.25 - 275) - 48.92$

$= 30.55985 - 48.92$

$\boxed{Q = -18.36\ kJ}$